Nirma University

Institute of Technology

Department of Computer Science & Engineering

Explainable AI

Lab 3: GLM, GAM, PDP, SHAP & ICE Plots

(Generalized Linear & Additive Models with Post-hoc Explanations)

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1 Objective

This lab explores advanced interpretable models and visualization techniques for explainability:

1. **Generalized Linear Models (GLM)** — extending linear regression with non-Gaussian distributions
2. **Generalized Additive Models (GAM)** — capturing non-linear feature effects with splines
3. **Partial Dependence Plots (PDP)** — visualizing marginal feature effects
4. **SHAP Values** — computing Shapley-based feature attributions for GAMs
5. **Individual Conditional Expectation (ICE) Plots** — revealing heterogeneous feature effects

2 Datasets

2.1 Diabetes Dataset

- **Source:** `sklearn.datasets.load_diabetes`
- **Task:** Regression (predicting disease progression)
- **Samples:** 442
- **Features (10):** age, sex, bmi, bp, s1–s6 (blood serum measurements)
- **Target:** Quantitative measure of diabetes progression one year after baseline
- **Split:** 80/20 train-test split (`random_state=42`)

2.2 Breast Cancer Wisconsin Dataset

- **Source:** `sklearn.datasets.load_breast_cancer`
- **Task:** Binary classification (Malignant vs. Benign)
- **Samples:** 569 (212 malignant, 357 benign)
- **Features Used (10 of 30):** First 10 mean features (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension)
- **Target:** 0 = Malignant, 1 = Benign
- **Split:** 80/20 train-test split

3 Methodology

3.1 Generalized Linear Models (GLM)

3.1.1 Theory

A GLM extends ordinary linear regression by allowing the response variable to follow any distribution from the exponential family. A GLM consists of three components:

- **Random component:** The distribution of Y (e.g., Gaussian, Gamma, Binomial)
- **Systematic component:** The linear predictor $\eta = \mathbf{X}\boldsymbol{\beta}$
- **Link function:** $g(\mu) = \eta$, connecting the mean response to the linear predictor

$$g(E[Y \mid \mathbf{X}]) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p \tag{1}$$

3.1.2 GLM for Diabetes (Regression)

The diabetes target is continuous, positive, and right-skewed. Therefore:

- **Family:** Gamma (appropriate for positive, skewed data)
- **Link:** Log (ensures predictions remain positive)

With the log link, coefficients have a **multiplicative interpretation**: e^{β_j} gives the multiplicative change in the expected response for a unit increase in x_j.

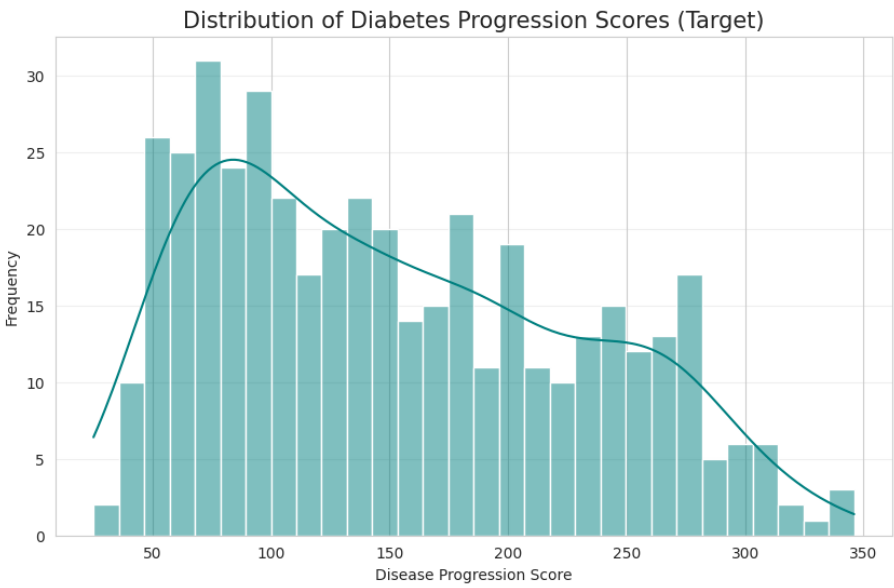

Figure 1: Distribution of diabetes target variable. The right-skewed, positive-valued distribution motivates the use of a Gamma GLM with log link.

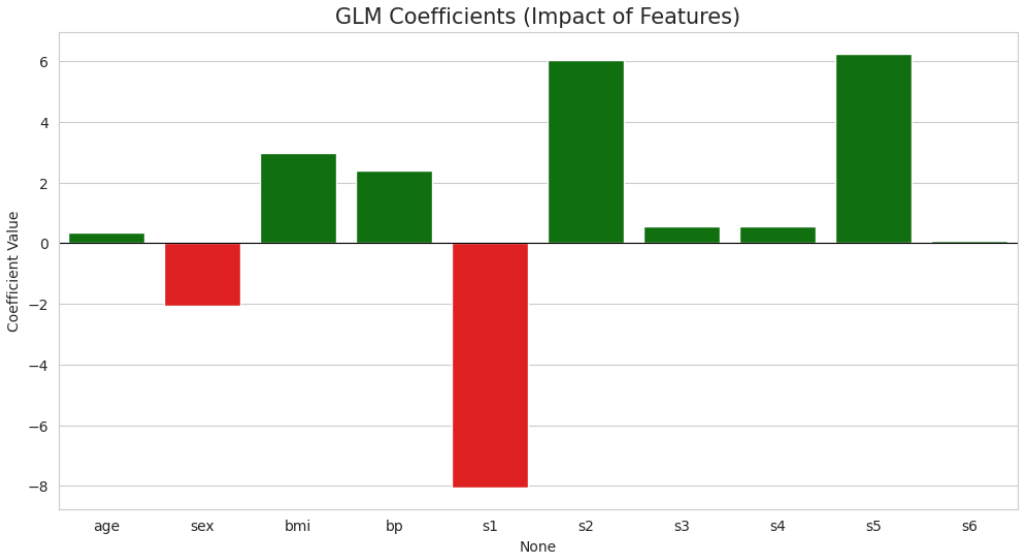

Figure 2: GLM coefficients for diabetes prediction. Green bars indicate positive effects (increase disease progression); red bars indicate negative effects.

3.1.3 GLM for Breast Cancer (Classification)

For binary classification, the appropriate choice is:

- **Family:** Binomial
- **Link:** Logit (maps probabilities to $(-\infty, +\infty)$)

$$\text{logit}(P(Y=1)) = \log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \beta_0 + \sum_{j=1}^{p} \beta_j x_j \tag{2}$$

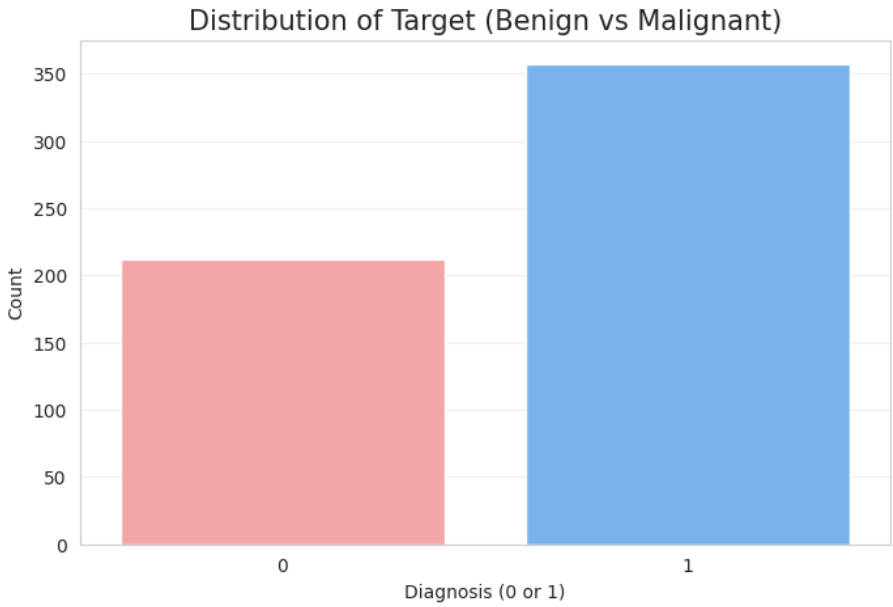

Figure 3: Distribution of breast cancer target variable (binary). The imbalanced distribution justifies using Binomial GLM with Logit link.

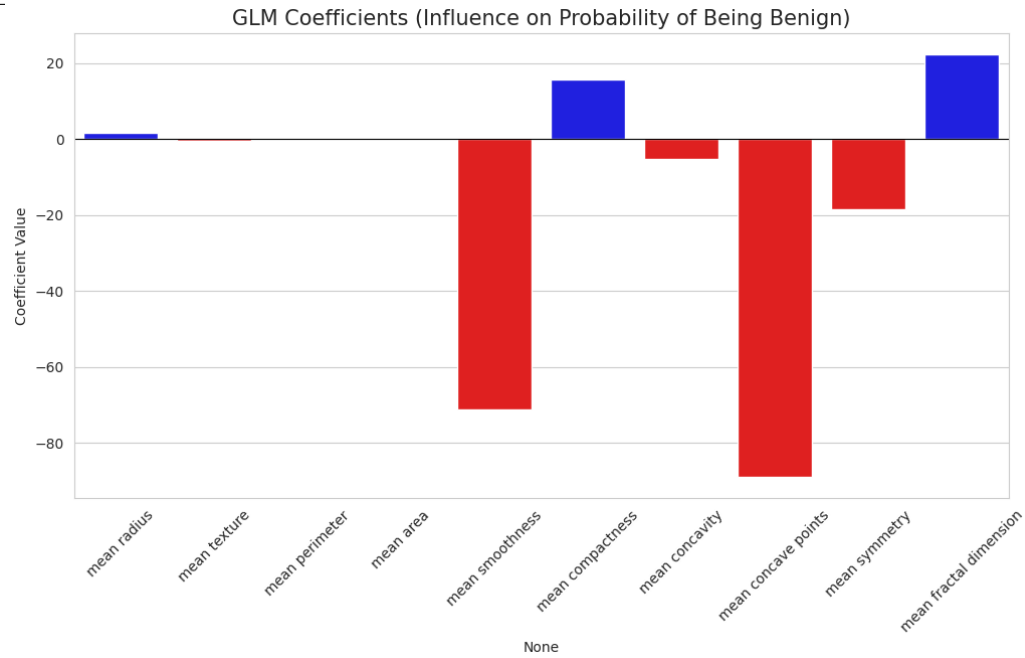

Figure 4: GLM coefficients for breast cancer classification. Positive coefficients (blue) increase the probability of benign diagnosis; negative coefficients (red) indicate higher malignancy risk.

3.2 Generalized Additive Models (GAM)

3.2.1 Theory

GAMs extend GLMs by replacing linear terms with smooth, non-linear functions:

$$g(E[Y]) = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_p(x_p) \tag{3}$$

Each f_j is typically a spline function that can capture non-linear relationships. The **Effective Degrees of Freedom (EDoF)** of each spline indicates the complexity of the learned shape:

- EDoF ≈ 1: Nearly linear relationship
- EDoF > 3: Complex, non-linear relationship

3.2.2 Implementation

- **Diabetes:** `LinearGAM` with 10 spline terms, optimized via `gridsearch()`
- **Results:** MSE $= 2750.61$, $R^2 = 0.48$
- **Cancer:** `LogisticGAM` with 10 spline terms
- **Results:** Accuracy $= 96.49\%$, Confusion Matrix: $[[41, 2], [2, 69]]$

3.3 Partial Dependence Plots (PDP)

3.3.1 Theory

PDPs show the marginal effect of a feature x_j on the predicted outcome, averaging over all other features:

$$\hat{f}_j(x_j) = \frac{1}{n} \sum_{i=1}^{n} f(x_j, \mathbf{x}_{i,-j}) \tag{4}$$

This reveals the "average" relationship between a feature and the prediction.

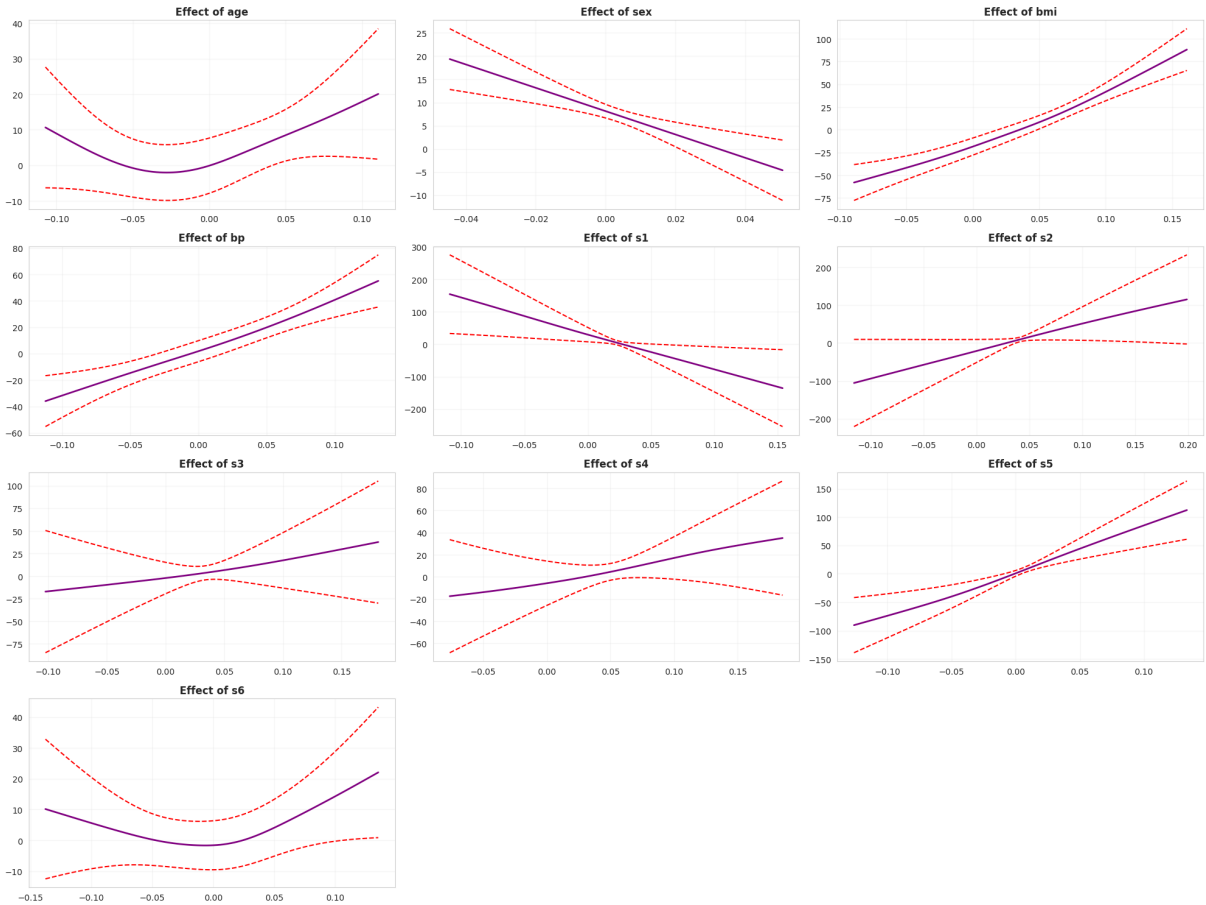

Figure 5: Partial Dependence Plots for all 10 features in the diabetes GAM. Purple lines show the marginal effect; dashed lines show 95% confidence intervals. Features like `bmi` show strong non-linear effects.

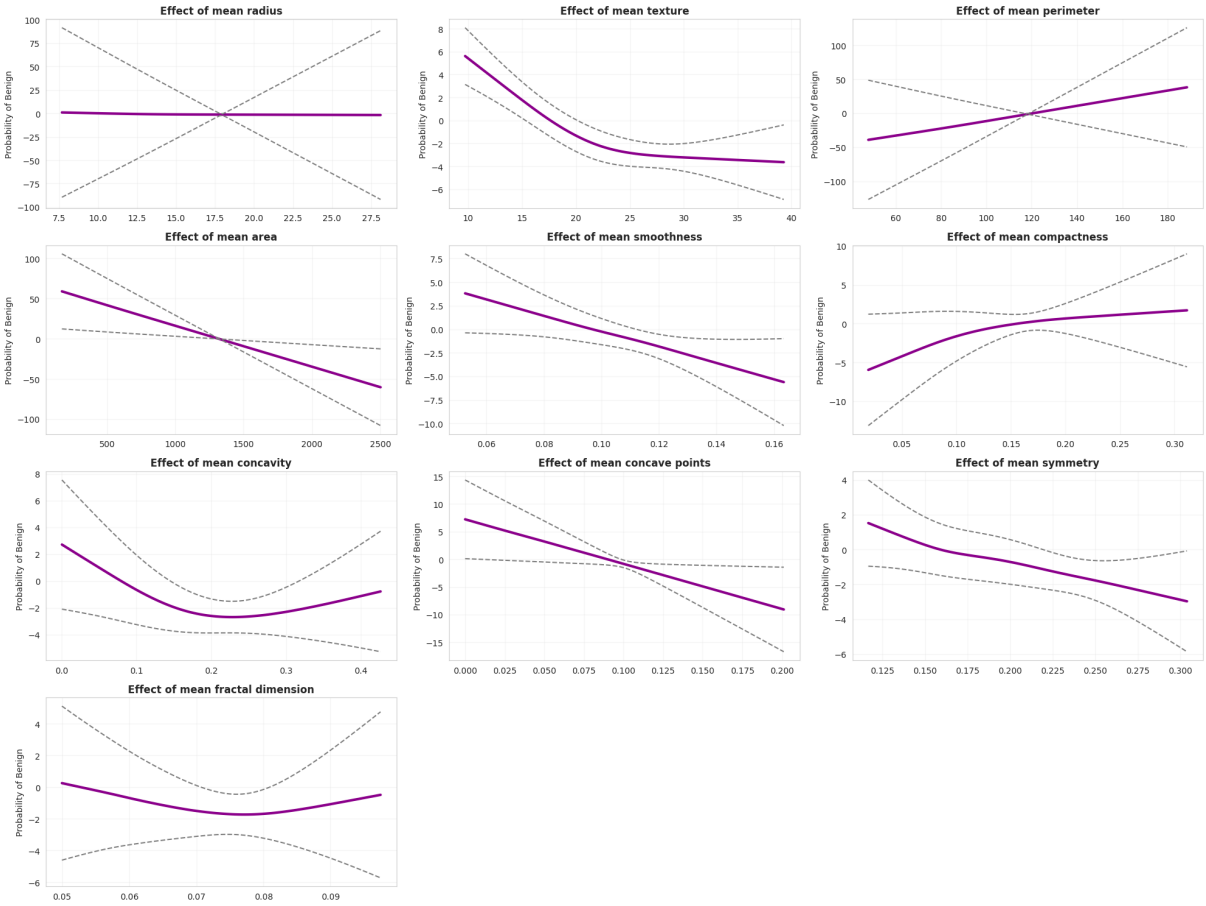

Figure 6: Partial Dependence Plots for 10 breast cancer features from LogisticGAM. The y-axis shows the probability of a benign diagnosis. Mean radius shows a dramatic negative effect.

3.4 SHAP Analysis on GAM

3.4.1 Theory

SHAP values decompose a prediction into additive feature contributions based on cooperative game theory:

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[f(S \cup \{j\}) - f(S) \right] \tag{5}$$

Since GAMs are not tree-based, `KernelExplainer` from SHAP is used, which works with any model by perturbing inputs and observing predictions.

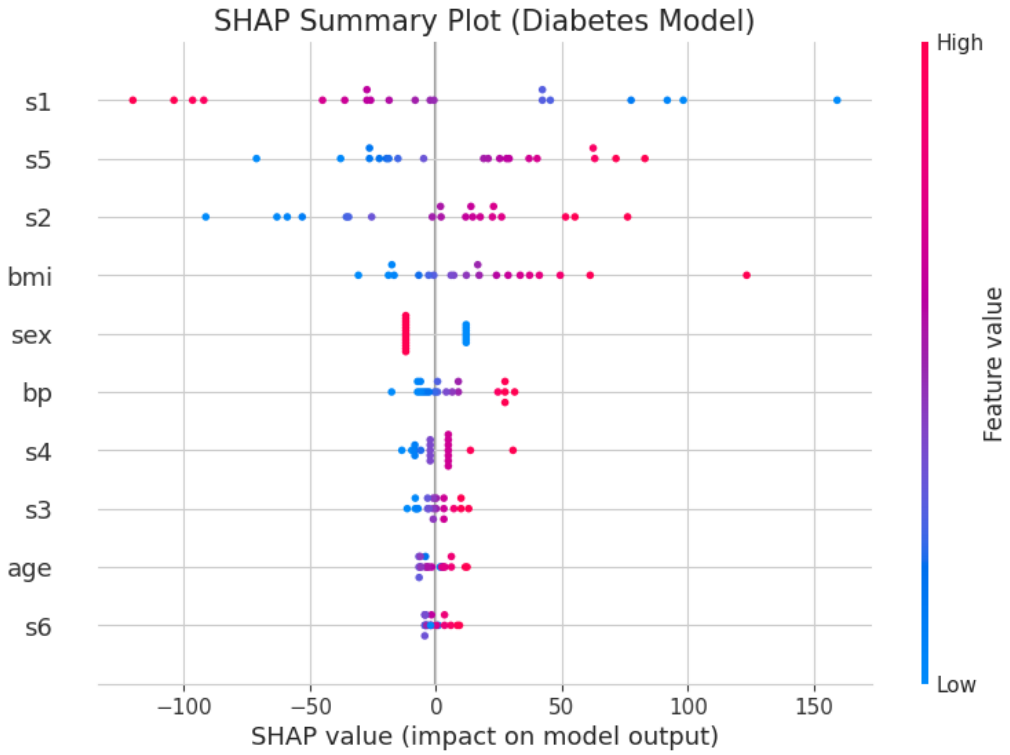

Figure 7: SHAP summary plot for the diabetes GAM model. Features `bmi` and `s5` show the widest SHAP value ranges, confirming their importance.

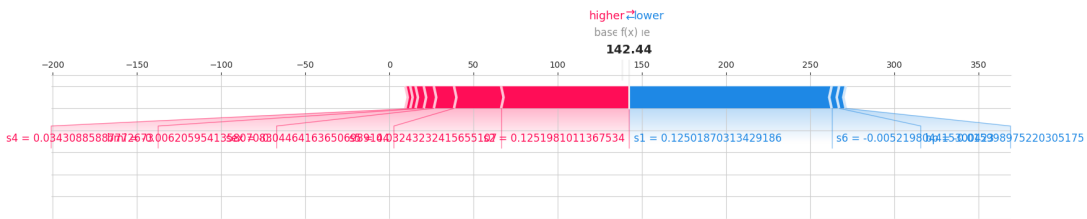

Figure 8: SHAP force plot for a single patient prediction, showing how each feature pushes the prediction up or down from the base value.

3.5 Individual Conditional Expectation (ICE) Plots

3.5.1 Theory

ICE plots extend PDPs by showing the feature-response curve for **each individual observation**, rather than just the average:

$$\hat{f}_j^{(i)}(x_j) = f(x_j, \mathbf{x}_{i,-j}) \tag{6}$$

Centered ICE (c-ICE) plots subtract the value at the minimum point to highlight differences in slopes:

$$\hat{f}_{j,\text{centered}}^{(i)}(x_j) = \hat{f}_j^{(i)}(x_j) - \hat{f}_j^{(i)}(x_{j,\min}) \tag{7}$$

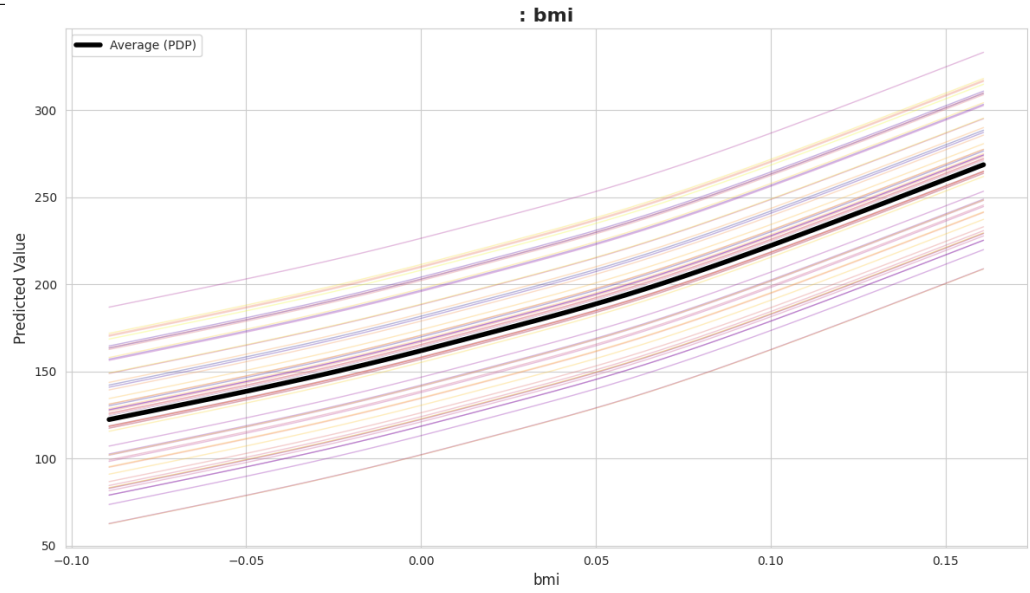

Figure 9: ICE plots for key diabetes features. Each thin line represents one patient; the thick line is the average (PDP). The spread of lines reveals heterogeneous feature effects.

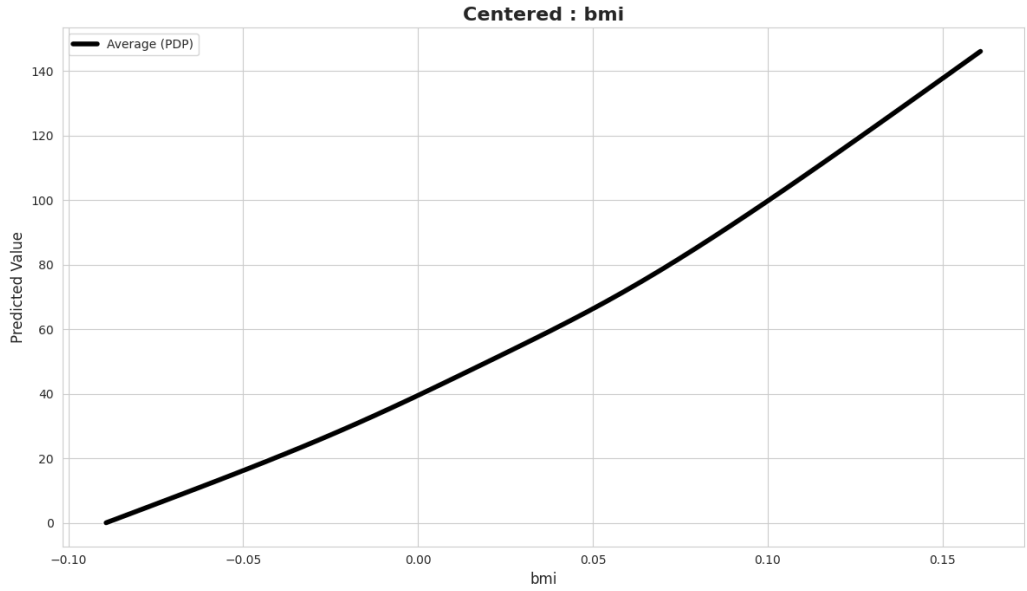

Figure 10: Centered ICE (c-ICE) plots. All curves are anchored at zero, making it easier to see differences in how individual patients respond to feature changes.

4 Results and Discussion

- **GLM:** The Gamma GLM with log link identified `bmi` and `s5` as the strongest predictors of diabetes progression. The Binomial GLM correctly identified `mean radius` as the most critical feature for malignancy.
- **GAM:** By allowing non-linear feature shapes, GAMs captured complex relationships that linear models miss. The cancer GAM achieved 96.49% accuracy with

only 4 misclassifications.

- **PDP:** Partial dependence plots revealed that features like `bmi` have a non-linear, accelerating effect on diabetes progression — information lost in linear coefficient summaries.
- **SHAP:** Confirmed that `bmi` and `s5` dominate predictions, while providing individual-level explanations through force plots.
- **ICE:** Revealed that patients respond differently to the same feature changes — some show steep increases in risk with higher `bmi`, while others are relatively unaffected. This heterogeneity is hidden in averaged PDPs.

5 Conclusion

This lab demonstrated a progression of explainability techniques from simple coefficient analysis (GLM) to rich, individualized explanations (ICE plots and SHAP). Key takeaways:

- GLMs provide a principled extension of linear regression with interpretable coefficients
- GAMs capture non-linear effects while maintaining additive interpretability
- PDPs summarize the average effect of each feature, while ICE plots reveal individual variation
- SHAP values provide a theoretically rigorous framework for both global and local explanations